

ANDREW (KOHLER) WOOD III

MECHANICAL ENGINEER, EIT

Entry-level mechanical engineer with hands-on experience taking CAD designs from concept through prototype and fabrication.

CONTACT

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📍 Lakewood, CO

CERTIFICATIONS

Engineer in Training EIT

Received March 2026

FE Mechanical Exam

Passed October 2025

Six Sigma Yellow Belt

Awarded by Colorado School of Mines May 2025

SKILLS

CAD

SolidWorks parts/assemblies/drawings, GD&T basic, tolerance stack-ups basic, DFM/DFA

Simulation

FEA linear static, buckling, CFD SolidWorks Flow Simulation—basic internal flow/heat transfer

Manufacturing

CNC router/mill/laser/waterjet/plasma, MIG welding, metal/wood fabrication, FDM/SLA 3D printing

Methods/Tools

FMEA, Scrum sprints, backlog, standups

Programming

MATLAB, Python, C++, Arduino, LaTeX

EXTRACURRICULAR & INTERESTS

ASME member · Woodworking · Certified Rescue Diver · Advanced skier · Jewelry fabrication

EXPERIENCE

Prototyping & Fabrication Projects

March 2025 – Present

LABRIOLA INNOVATION HUB, MINES • GOLDEN, CO

- Designed and fabricated woodworking items cutting boards, rolling pins, record cabinet using SolidWorks models/drawings; executed long miter joints requiring tight fit-up and consistent tolerances.
- Operated ShopBot CNC router and shop equipment; selected grain orientation and specified food-safe adhesives/finishes; improved dimensional consistency through refined setups.

Senior Capstone Project – Adaptive Bass Guitar

Aug 2024 – May 2025

COLORADO SCHOOL OF MINES • GOLDEN, CO

- Redesigned the plucking mechanism to eliminate failure modes from flexible/unstable connections; implemented 3D-printed paddle interfaces to improve reliability in repeated use.
- Designed a player stabilization system shoulder strap + hip strap to counteract playing-induced torque and replace a tripod-based neck support, improving usability and reducing setup inconvenience.
- Modeled assemblies/mechanisms in SolidWorks and produced a detailed assembly manual.
- Ran work in 2-week Scrum sprints; tracked risks/mitigations with FMEA.

Mechanical Design Assistant

Summer 2022

JG WORKS • HAILEY, ID

- Fabricated mono-ski hardware for Special Olympics athletes.
- Assisted in designing modular office space using 3-axis CNC cut plywood sheets.

Fabrication Intern

Summer 2021

PHRED'S FABRICATION • KETCHUM, ID

- Assisted with layout, fabrication, and installation of steel railings for residential projects.

EDUCATION

Colorado School of Mines

Aug 2021 – Aug 2025

B.S. Mechanical Engineering
GOLDEN, CO

Macy-Naszkowski School of Art Music

Oct 2022 – Present

Additional Training: Classical Piano
LAKEWOOD, CO

